Execution Cost Frontier

Speed vs. slippage trade-off

PROFESSIONAL DEFINITION

The Execution Cost Frontier expresses the trade-off between trading speed and market impact (slippage): trading faster moves the price against you, while trading slowly risks adverse moves.

WHO DEVELOPED IT

Grounded in optimal-execution research, notably the Almgren-Chriss framework.

WHY IT WAS DEVELOPED

It was developed to optimise trade execution by balancing the cost of speed against the risk of delay.

FIGURE 1 · EXECUTION COST FRONTIER

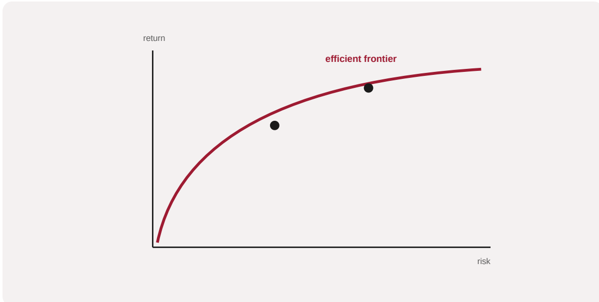

Figure 1. A schematic of the Execution Cost Frontier as applied in BARBM312 (illustrative).

HOW IT IS USED

Execution strategies are placed on the frontier; the schedule is chosen to minimise total expected cost given risk preferences.

WHEN IT IS USED

When executing large trades and designing execution algorithms.

BY WHOM IT IS USED

Traders, execution teams and quantitative researchers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It formalises the speed-versus-impact trade-off, underpinning algorithmic execution strategy.

RELATED FRAMEWORKS IN THIS COURSE

Algo-Trading Pipeline · Slibrary 24 | **Latency Arms Race** · Slibrary 24 | **Market Microstructure Map** · Slibrary 24 | **Flash-Crash Risk Loop** · Slibrary 24

SOURCES (HARVARD REFERENCING STYLE)

Almgren, R. and Chriss, N. (2001) 'Optimal execution of portfolio transactions', Journal of Risk, 3(2).

Kissell, R. (2013) The Science of Algorithmic Trading and Portfolio Management. London: Academic Press.